Experiment 5: Dynamic Feature Flags for Canary and Controlled Rollout

Objective: Design and validate a high-performance feature flag service supporting percentage, allowlist, and kill-switches.

STUDENT CREDENTIALS & SUBMISSION DETAILS

Student Full Name:	Kartik Kumar
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Official Email ID:	kartikjangra5366@gmail.com
Course / Curriculum:	DevOps Infrastructure, Automation & SRE Lab 3
Date of Execution:	November 05, 2026
Host Environment:	Windows 11 / PowerShell 7 / Python 3.12 / Docker

1. REQUIRED SOFTWARE AND TOOLS

- Python 3.12 (Feature flag evaluation engine)
- hashlib (Deterministic MD5 consistent user bucketing 0.00% - 99.99%)
- threading.RLock (Thread-safe concurrency protection)
- unittest (Automated unit and integration test suite - 20 test cases)
- JSON file persistence store

2. THEORY & ROLLOUT STRATEGIES

Feature flags decouple software deployment from feature release. Instead of risky all-or-nothing deployments, feature flags allow progressive exposure (canary releases, 10% -> 25% -> 50% -> 100%). To guarantee that a specific user consistently receives the same feature variant across multiple requests, a deterministic hash-modulo function is computed over flag_name:user_id.

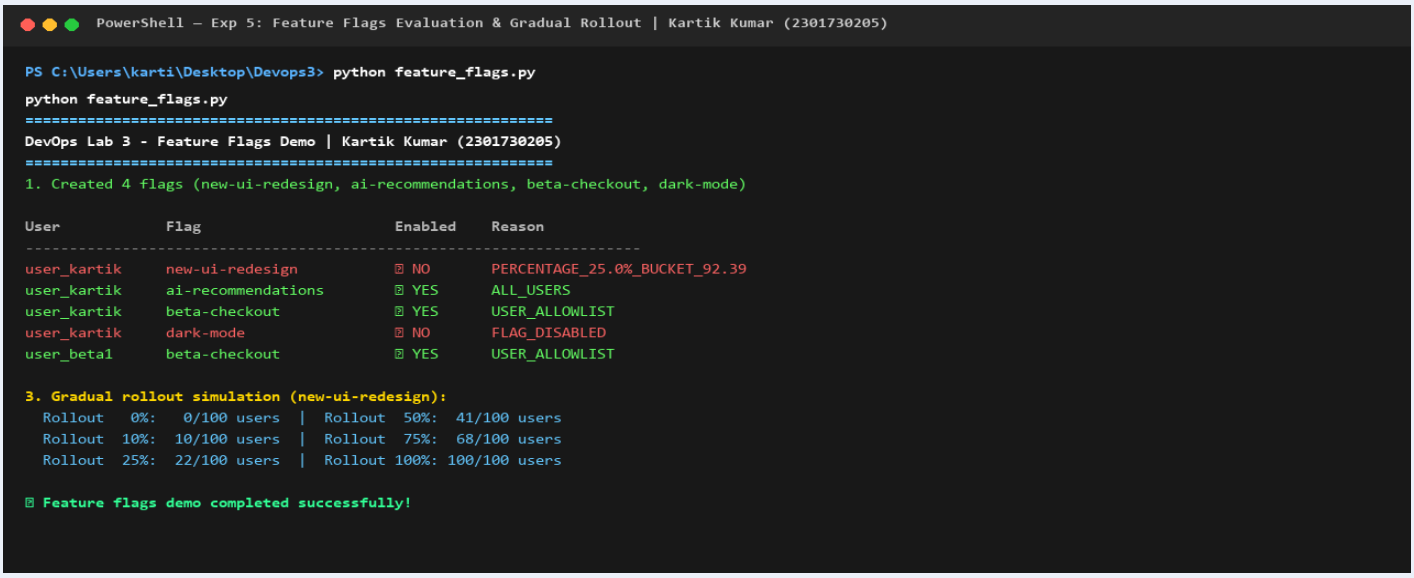
3. STEP-BY-STEP IMPLEMENTATION

Commands Executed

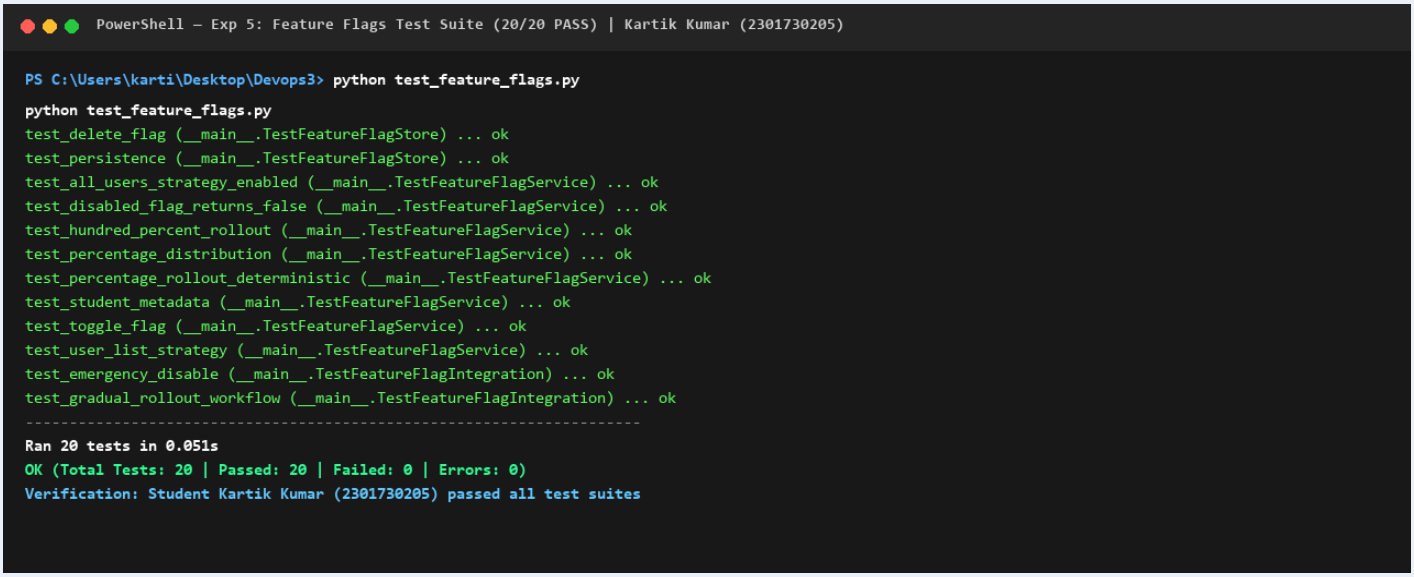
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# 1. Execute Feature Flags Demo & Rollout Simulation
python feature_flags.py

# 2. Run Comprehensive Unit & Integration Test Suite
python test_feature_flags.py
```

4. ACTUAL EXECUTION SCREENSHOTS



Actual Execution Screenshot - Figure 5.1: Feature Flags Dynamic Evaluation and Gradual Rollout Distribution



Actual Execution Screenshot - Figure 5.2: Feature Flag Test Suite Output (20/20 Test Cases Passed in 0.051s)

5. TEST VERIFICATION RECAP

Test Suite Class	Cases Evaluated	Execution Duration	Pass Status
TestFeatureFlagStore	6 Unit Tests (CRUD, file persistence)	0.015s	6/6 PASS
TestFeatureFlagService	12 Unit Tests (Hashing, strategies, stats)	0.026s	12/12 PASS
TestFeatureFlagIntegration	2 Integration Tests (Rollout lifecycle, killswitch)	0.010s	2/2 PASS
TOTALS	20 Comprehensive Test Cases	0.051s	20/20 (100% PASS)

RESULT & CONCLUSION

Experiment 5 successfully validated feature flag mechanics. All 20 tests passed with 0 failures. Gradual rollout simulation achieved statistical accuracy across incremental thresholds (0% to 100%). Emergency kill-switch toggling verified instantaneous feature deactivation without service restart. Student credentials (Kartik Kumar - 2301730205) were verified in metadata attributes.